

Cleaning Data and Creating Main Dataset

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Libraries

```
library(dplyr)
library(lubridate)
library(tidyr)
```

Population Data

```
# load data
pop <- read.csv("Datasets/scPopulation.csv", header = T)

# make the data match other datasets
pop.expanded <- pop %>%
  rowwise() %>%
  mutate(Date = list(seq(ymd(paste0(Year, "-01-01")), ymd(paste0(Year, "-12-01")), by = "month") %>%
    ceiling_date("month") - days(1))) %>%
  unnest(Date) %>%
  select(Date, Population) %>%
  arrange(Date)

# make population an integer column
pop.expanded$Population <- gsub(",", "", pop.expanded$Population) # Remove commas
pop.expanded$Population <- as.numeric(pop.expanded$Population)

# check data
print(pop.expanded)
```

```
## # A tibble: 648 x 2
##   Date      Population
##   <date>      <dbl>
## 1 1971-01-31  2662269
## 2 1971-02-28  2662269
## 3 1971-03-31  2662269
## 4 1971-04-30  2662269
```

```
## 5 1971-05-31 2662269
## 6 1971-06-30 2662269
## 7 1971-07-31 2662269
## 8 1971-08-31 2662269
## 9 1971-09-30 2662269
## 10 1971-10-31 2662269
## # i 638 more rows
```

UI Claims Data

Overview

Claims Data is gathered from the ETA 5159 report from SCDEW. I've used c1 as the column for initial claims in the past but I believe c2-c7 is the better choice. From this data, we will create a lagged claims column and a lagged claims percentage change column. The data originates in 1971.

Clean Claims Data

```
# gather data
claims <- read.csv("Datasets/ClaimsData/ar5159.csv", header = T)

# fix date column
colnames(claims)[colnames(claims) == "rptdate"] <- "Date"
claims$Date <- as.Date(claims$Date, format = "%m/%d/%Y")

# get rid of all states but SC
claims <- filter(claims, st=="SC")

# join population data
claims <- claims %>%
  left_join(pop.expanded, by = "Date")

# combine c2-c7 and drop extra columns
claims <- claims %>%
  mutate(InitialClaims = c2 + c3 + c4 + c5 + c6 + c7) %>%
  select(Date, Population, InitialClaims)

# create Initial Claims per Capita
claims <- claims %>%
  mutate(InitialClaims.PerCapita = InitialClaims / Population)

# create Initial Claims % change
claims <- claims %>%
  arrange(Date) %>% # check for chronological order
  mutate(InitialClaims.PerChange = (InitialClaims / lag(InitialClaims) - 1) * 100)

# create lagged claims and lagged claims % change
claims <- claims %>%
```

```

arrange(Date) %>% # check for chronological order
mutate(InitialClaims.Lag = lead(InitialClaims),
       InitialClaims.Lag.PerCapita = lead(InitialClaims.PerCapita),
       InitialClaims.Lag.PerChange = (lead(InitialClaims) / InitialClaims - 1) * 100)

# create lagged 2 month claims and lagged 2 month claims % change
claims <- claims %>%
  arrange(Date) %>% # check for chronological order
  mutate(InitialClaims.Lag2 = lead(InitialClaims, 2),
         InitialClaims.Lag2.PerCapita = lead(InitialClaims.PerCapita, 2),
         InitialClaims.Lag2.PerChange = (lead(InitialClaims, 2) / InitialClaims.Lag - 1) * 100)

# check data
tail(claims, 5)

```

```

##           Date Population InitialClaims InitialClaims.PerCapita
## 646 2024-10-31   5478831         11596         0.002116510
## 647 2024-11-30   5478831          7785         0.001420924
## 648 2024-12-31   5478831         13464         0.002457459
## 649 2025-01-31         NA         14667             NA
## 650 2025-02-28         NA          8750             NA
##      InitialClaims.PerChange InitialClaims.Lag InitialClaims.Lag.PerCapita
## 646              54.202128              7785         0.001420924
## 647             -32.864781             13464         0.002457459
## 648              72.947977             14667             NA
## 649              8.934938              8750             NA
## 650             -40.342265               NA             NA
##      InitialClaims.Lag.PerChange InitialClaims.Lag2 InitialClaims.Lag2.PerCapita
## 646             -32.864781             13464         0.002457459
## 647              72.947977             14667             NA
## 648              8.934938              8750             NA
## 649             -40.342265               NA             NA
## 650              NA               NA             NA
##      InitialClaims.Lag2.PerChange
## 646              72.947977
## 647              8.934938
## 648             -40.342265
## 649              NA
## 650              NA

```

Laus Data

Overview

This data is gathered from LAUS and provides employment, unemployment and labor force numbers in SC from 1976.

Load in Data

```
# gather data
unemployment <- read.csv("Datasets/Laus/UnemploymentSC.csv", header=T)
employment <- read.csv("Datasets/Laus/EmploymentSC.csv", header=T)
laborforce <- read.csv("Datasets/Laus/LaborForceSC.csv", header=T)

# rename "Value"
unemployment <- unemployment %>%
  rename(Unemployment = Value)

employment <- employment %>%
  rename(Employment = Value)

laborforce <- laborforce %>%
  rename(LaborForce = Value)
```

Fix Date Columns

```
# list of all months
month_map <- c("M01" = 1, "M02" = 2, "M03" = 3, "M04" = 4, "M05" = 5, "M06" = 6,
              "M07" = 7, "M08" = 8, "M09" = 9, "M10" = 10, "M11" = 11, "M12" = 12)

# convert months to a number
unemployment$Month <- month_map[unemployment$Period]
employment$Month <- month_map[employment$Period]
laborforce$Month <- month_map[laborforce$Period]

# create date column
unemployment$Date <- as.Date(paste(unemployment$Year, unemployment$Month, "01", sep = "-"))
employment$Date <- as.Date(paste(employment$Year, employment$Month, "01", sep = "-"))
laborforce$Date <- as.Date(paste(laborforce$Year, laborforce$Month, "01", sep = "-"))

# make date be the last day of the month
unemployment$Date <- ceiling_date(unemployment$Date, "month") - days(1)
employment$Date <- ceiling_date(employment$Date, "month") - days(1)
laborforce$Date <- ceiling_date(laborforce$Date, "month") - days(1)
```

Create Percentage Change Columns

```
# add new % change column to every df individually
unemployment <- unemployment %>%
  arrange(Date) %>% # check for chronological order
  mutate(Unemployment.Per = (Unemployment / lag(Unemployment) - 1) * 100)

employment <- employment %>%
  arrange(Date) %>% # check for chronological order
  mutate(Employment.Per = (Employment / lag(Employment) - 1) * 100)
```

```
laborforce <- laborforce %>%
  arrange(Date) %>% # check for chronological order
  mutate(LaborForce.Per = (LaborForce / lag(LaborForce) - 1) * 100)
```

Combine Datasets

```
# create Laus dataset with all numbers
Laus <- unemployment %>%
  select(Date, Unemployment, Unemployment.Per) %>% # we only want to keep these two columns
  full_join(employment %>% select(Date, Employment, Employment.Per), by = "Date") %>%
  full_join(laborforce %>% select(Date, LaborForce, LaborForce.Per), by = "Date")

# create unemployment rate
Laus <- Laus %>%
  mutate(UnemploymentRate = Unemployment / LaborForce * 100)

# create unemployment rate percentage change
Laus <- Laus %>%
  arrange(Date) %>% # check for chronological order
  mutate(UnemploymentRate.Per = (UnemploymentRate / lag(UnemploymentRate) - 1) * 100)
```

Make Per Capita Columns

```
# join laus data with population data
Laus <- Laus %>%
  left_join(pop.expanded, by = "Date")

# create per capita columns for Unemployment, Employment, and Labor Force
Laus <- Laus %>%
  mutate(Unemployment.PerCapita = Unemployment / Population,
         Employment.PerCapita = Employment / Population,
         LaborForce.PerCapita = LaborForce / Population)

# keep desired columns and organize
Laus <- Laus %>%
  select(Date, Unemployment, Unemployment.Per, Unemployment.PerCapita,
         Employment, Employment.Per, Employment.PerCapita,
         LaborForce, LaborForce.Per, LaborForce.PerCapita,
         UnemploymentRate, UnemploymentRate.Per)

# check Laus data
tail(Laus, 5)
```

```
##           Date Unemployment Unemployment.Per Unemployment.PerCapita Employment
## 584 2024-08-31      107891      10.358619      0.01969234      2414624
## 585 2024-09-30      113975       5.639025      0.02080280      2413300
## 586 2024-10-31      118097       3.616583      0.02155515      2413165
## 587 2024-11-30      121482       2.866288      0.02217298      2412589
## 588 2024-12-31      119485      -1.643865      0.02180848      2417169
```

	Employment.Per	Employment.PerCapita	LaborForce	LaborForce.Per
## 584	-0.02211035	0.4407188	2522515	0.3817468
## 585	-0.05483255	0.4404772	2527275	0.1887006
## 586	-0.00559400	0.4404525	2531262	0.1577589
## 587	-0.02386907	0.4403474	2534071	0.1109723
## 588	0.18983756	0.4411833	2536654	0.1019308

	LaborForce.PerCapita	UnemploymentRate	UnemploymentRate.Per
## 584	0.4604112	4.277120	9.938930
## 585	0.4612800	4.509798	5.440059
## 586	0.4620077	4.665538	3.453376
## 587	0.4625204	4.793946	2.752261
## 588	0.4629918	4.710339	-1.744018

SP500 Data

Overview

This data involves SP500 data, and some of its indices, from the WSJ. It will put in a monthly format where the value is the final close value from the last trading day of each month. However, we will change all dates to be the last day of the month, even if the last trading day isn't the last day of the month, so it can be joined with other datasets.

Clean SP500 Dataset

```
# gather data
sp500 <- read.csv("Datasets/SP500/sp500.csv") %>%
  mutate(Date = mdy(Date))

# last trading day of each month
sp500_last_trading <- sp500 %>%
  group_by(YearMonth = floor_date(Date, "month")) %>%
  slice_max(Date) %>%
  ungroup()

# dataframe with the actual last day of each month
sp500_monthly <- sp500_last_trading %>%
  mutate(Date = ceiling_date(YearMonth, "month") - days(1)) %>%
  select(-YearMonth) # Remove helper column

# drop extra columns
sp500_monthly <- sp500_monthly %>%
  select(-Open, -High, -Low)

# create percentage change column
sp500_monthly <- sp500_monthly %>%
  arrange(Date) %>% # check for chronological order
  mutate(SP500.Per = (Close / lag(Close) - 1) * 100)
```

```
# rename Close
sp500_monthly <- sp500_monthly %>%
  rename(SP500 = Close)
```

Clean SP500 Healthcare Index

```
# gather data
sp500.health <- read.csv("Datasets/SP500/sp500_healthcare.csv") %>%
  mutate(Date = mdy(Date))

# last trading day of each month
sp500.health_last_trading <- sp500.health %>%
  group_by(YearMonth = floor_date(Date, "month")) %>%
  slice_max(Date) %>%
  ungroup()

# dataframe with the actual last day of each month
sp500.health_monthly <- sp500.health_last_trading %>%
  mutate(Date = ceiling_date(YearMonth, "month") - days(1)) %>%
  select(-YearMonth) # Remove helper column

# drop extra columns
sp500.health_monthly <- sp500.health_monthly %>%
  select(-Open, -High, -Low)

# create percentage change column
sp500.health_monthly <- sp500.health_monthly %>%
  arrange(Date) %>% # check for chronological order
  mutate(SP500_Health.Per = (Close / lag(Close) - 1) * 100)

# rename Close
sp500.health_monthly <- sp500.health_monthly %>%
  rename(SP500_Health = Close)
```

Clean SP500 Energy Index

```
# gather data
sp500.energy <- read.csv("Datasets/SP500/sp500_energy.csv") %>%
  mutate(Date = mdy(Date))

# last trading day of each month
sp500.energy_last_trading <- sp500.energy %>%
  group_by(YearMonth = floor_date(Date, "month")) %>%
  slice_max(Date) %>%
  ungroup()

# dataframe with the actual last day of each month
sp500.energy_monthly <- sp500.energy_last_trading %>%
  mutate(Date = ceiling_date(YearMonth, "month") - days(1)) %>%
```

```

select(-YearMonth) # Remove helper column

# drop extra columns
sp500.energy_monthly <- sp500.energy_monthly %>%
  select(-Open, -High, -Low)

# create percentage change column
sp500.energy_monthly <- sp500.energy_monthly %>%
  arrange(Date) %>% # check for chronological order
  mutate(SP500_Energy.Per = (Close / lag(Close) - 1) * 100)

# rename Close
sp500.energy_monthly <- sp500.energy_monthly %>%
  rename(SP500_Energy = Close)

```

Combine DF's into one main DF

```

# join df's by date
SP500.FINAL <- sp500_monthly %>%
  full_join(sp500.health_monthly, by = "Date") %>%
  full_join(sp500.energy_monthly, by = "Date")

# check values
head(SP500.FINAL, 5)

```

```

## # A tibble: 5 x 7
##   Date      SP500 SP500.Per SP500_Health SP500_Health.Per SP500_Energy
##   <date>    <dbl>    <dbl>         <dbl>         <dbl>         <dbl>
## 1 1981-01-31  130.      NA             NA             NA             NA
## 2 1981-02-28  131.     1.33          NA             NA             NA
## 3 1981-03-31  136     3.60          NA             NA             NA
## 4 1981-04-30  133.    -2.30          NA             NA             NA
## 5 1981-05-31  133.    -0.211         NA             NA             NA
## # i 1 more variable: SP500_Energy.Per <dbl>

```

FRED DATA

Overview

This is data from FRED that gives us various job statistics in the US.

Clean FRED Data


```

# gather data
openings <- read.csv("Datasets/FRED-Jobs/openings.csv", header=T)
hires <- read.csv("Datasets/FRED-Jobs/hires.csv", header=T)
separations <- read.csv("Datasets/FRED-Jobs/separations.csv", header=T)
quits <- read.csv("Datasets/FRED-Jobs/quits.csv", header=T)
layoffs <- read.csv("Datasets/FRED-Jobs/layoffs.csv", header=T)
lfpr <- read.csv("Datasets/FRED-Jobs/lfpr.csv", header=T)

# rename columns for each dataset, fix dates, and add percentage change
openings <- openings %>%
  rename(Openings = JTSJOL) %>%
  rename(Date = observation_date) %>%
  mutate(Date = as.Date(Date, format = "%Y-%m-%d"),
         Openings.Per = (Openings / lag(Openings) - 1) * 100)

hires <- hires %>%
  rename(Hires = JTSHIL) %>%
  rename(Date = observation_date) %>%
  mutate(Date = as.Date(Date, format = "%Y-%m-%d"),
         Hires.Per = (Hires / lag(Hires) - 1) * 100)

separations <- separations %>%
  rename(Separations = JTSTSL) %>%
  rename(Date = observation_date) %>%
  mutate(Date = as.Date(Date, format = "%Y-%m-%d"),
         Separations.Per = (Separations / lag(Separations) - 1) * 100)

quits <- quits %>%
  rename(Quits = JTSQUL) %>%
  rename(Date = observation_date) %>%
  mutate(Date = as.Date(Date, format = "%Y-%m-%d"),
         Quits.Per = (Quits / lag(Quits) - 1) * 100)

layoffs <- layoffs %>%
  rename(Layoffs = JTSLDL) %>%
  rename(Date = observation_date) %>%
  mutate(Date = as.Date(Date, format = "%Y-%m-%d"),
         Layoffs.Per = (Layoffs / lag(Layoffs) - 1) * 100)

lfpr <- lfpr %>%
  rename(LFPR = CIVPART) %>%
  rename(Date = observation_date) %>%
  mutate(Date = as.Date(Date, format = "%Y-%m-%d"),
         LFPR.Per = (LFPR / lag(LFPR) - 1) * 100)

# combine datasets
FRED <- lfpr %>%
  full_join(hires, by = "Date") %>%
  full_join(separations, by = "Date") %>%
  full_join(quits, by = "Date") %>%
  full_join(layoffs, by = "Date") %>%
  full_join(openings, by = "Date")

```

```
# fix date to move back to last day of the last month
FRED <- FRED %>%
  mutate(Date = Date - days(1))

# check data
tail(FRED, 5)
```

```
##           Date LFPR   LFPR.Per Hires  Hires.Per Separations Separations.Per
## 922 2024-09-30 62.5 -0.3189793 5350 -3.0621489      5285      1.69328459
## 923 2024-10-31 62.5  0.0000000 5307 -0.8037383      5087     -3.74645222
## 924 2024-11-30 62.5  0.0000000 5374  1.2624835      5082     -0.09828976
## 925 2024-12-31 62.6  0.1600000 5393  0.3535541      5252      3.34513971
## 926 2025-01-31 62.4 -0.3194888    NA         NA         NA         NA
##      Quits Quits.Per Layoffs Layoffs.Per Openings Openings.Per
## 922  3217  4.278768    1788  -3.036876    7615    7.208222
## 923  3032 -5.750699    1739  -2.740492    8031    5.462902
## 924  3095  2.077836    1669  -4.025302    7508   -6.512265
## 925  3266  5.525040    1635  -2.037148    7740    3.090037
## 926    NA      NA      NA         NA      NA      NA
```

Create Final Dataset

```
# Join all datasets to UI Claims Dataset (some data doesn't go that far back)
FinalData <- claims %>%
  left_join(Laus, by = "Date") %>%
  left_join(SP500.FINAL, by = "Date") %>%
  left_join(FRED, by = "Date")

# check data
tail(FinalData, 10)
```

```
##           Date Population InitialClaims InitialClaims.PerCapita
## 641 2024-05-31  5478831         9438      0.001722630
## 642 2024-06-30  5478831        10380      0.001894565
## 643 2024-07-31  5478831        13851      0.002528094
## 644 2024-08-31  5478831         9458      0.001726281
## 645 2024-09-30  5478831         7520      0.001372556
## 646 2024-10-31  5478831        11596      0.002116510
## 647 2024-11-30  5478831         7785      0.001420924
## 648 2024-12-31  5478831        13464      0.002457459
## 649 2025-01-31      NA        14667      NA
## 650 2025-02-28      NA         8750      NA
##      InitialClaims.PerChange InitialClaims.Lag InitialClaims.Lag.PerCapita
## 641          -8.679245         10380      0.001894565
## 642           9.980928         13851      0.002528094
## 643          33.439306          9458      0.001726281
## 644         -31.716122          7520      0.001372556
## 645         -20.490590         11596      0.002116510
```

##	646	54.202128	7785	0.001420924
##	647	-32.864781	13464	0.002457459
##	648	72.947977	14667	NA
##	649	8.934938	8750	NA
##	650	-40.342265	NA	NA
##	InitialClaims.Lag.PerChange InitialClaims.Lag2 InitialClaims.Lag2.PerCapita			
##	641	9.980928	13851	0.002528094
##	642	33.439306	9458	0.001726281
##	643	-31.716122	7520	0.001372556
##	644	-20.490590	11596	0.002116510
##	645	54.202128	7785	0.001420924
##	646	-32.864781	13464	0.002457459
##	647	72.947977	14667	NA
##	648	8.934938	8750	NA
##	649	-40.342265	NA	NA
##	650	NA	NA	NA
##	InitialClaims.Lag2.PerChange Unemployment Unemployment.Per			
##	641	33.439306	83589	4.771753
##	642	-31.716122	89718	7.332304
##	643	-20.490590	97764	8.968100
##	644	54.202128	107891	10.358619
##	645	-32.864781	113975	5.639025
##	646	72.947977	118097	3.616583
##	647	8.934938	121482	2.866288
##	648	-40.342265	119485	-1.643865
##	649	NA	NA	NA
##	650	NA	NA	NA
##	Unemployment.PerCapita Employment Employment.Per Employment.PerCapita			
##	641	0.01525672	2406512	0.08296885
##	642	0.01637539	2410159	0.15154714
##	643	0.01784395	2415158	0.20741370
##	644	0.01969234	2414624	-0.02211035
##	645	0.02080280	2413300	-0.05483255
##	646	0.02155515	2413165	-0.00559400
##	647	0.02217298	2412589	-0.02386907
##	648	0.02180848	2417169	0.18983756
##	649	NA	NA	NA
##	650	NA	NA	NA
##	LaborForce LaborForce.Per LaborForce.PerCapita UnemploymentRate			
##	641	2490101	0.2335468	0.4544949
##	642	2499877	0.3925945	0.4562793
##	643	2512922	0.5218257	0.4586603
##	644	2522515	0.3817468	0.4604112
##	645	2527275	0.1887006	0.4612800
##	646	2531262	0.1577589	0.4620077
##	647	2534071	0.1109723	0.4625204
##	648	2536654	0.1019308	0.4629918
##	649	NA	NA	NA
##	650	NA	NA	NA
##	UnemploymentRate.Per SP500 SP500.Per SP500_Health SP500_Health.Per			
##	641	4.527632	5277.51	4.8021224
##	642	6.912572	5460.48	3.4669759
##	643	8.402428	5522.30	1.1321349
##	644	9.938930	5648.40	2.2834688
##	645			1670.88
##	646			1700.33
##	647			1742.67
##	648			1829.71
##	649			
##	650			
##	641			2.2295098
##	642			1.7625443
##	643			2.4901049
##	644			4.9946347

```

## 645          5.440059 5762.48  2.0196870      1796.48      -1.8161348
## 646          3.453376 5705.45 -0.9896781      1711.50      -4.7303616
## 647          2.752261 6032.38  5.7301352      1713.71       0.1291265
## 648         -1.744018 5881.63 -2.4990137      1604.75      -6.3581353
## 649              NA 6040.53  2.7016320      1710.76       6.6060134
## 650              NA 5954.50 -1.4242128      1734.28       1.3748276
##      SP500_Energy SP500_Energy.Per LFPR    LFPR.Per Hires  Hires.Per Separations
## 641          708.04      -0.9665012 62.6  0.0000000   5088 -8.7026736      5064
## 642          698.23      -1.3855149 62.7  0.1597444   5451  7.1344340      5434
## 643          712.42       2.0322816 62.7  0.0000000   5230 -4.0543020      5170
## 644          695.86      -2.3244715 62.7  0.0000000   5519  5.5258126      5197
## 645          676.47      -2.7864800 62.5 -0.3189793   5350 -3.0621489      5285
## 646          681.25       0.7066093 62.5  0.0000000   5307 -0.8037383      5087
## 647          724.06       6.2840367 62.5  0.0000000   5374  1.2624835      5082
## 648          654.85      -9.5586001 62.6  0.1600000   5393  0.3535541      5252
## 649          667.95       2.0004581 62.4 -0.3194888    NA      NA      NA
## 650          689.88       3.2831799  NA      NA      NA      NA      NA
##      Separations.Per Quits Quits.Per Layoffs Layoffs.Per Openings Openings.Per
## 641      -4.70455401  3284 -2.116244   1467 -11.891892   7412  -6.189090
## 642       7.30647709  3390  3.227771   1745  18.950239   7504   1.241230
## 643      -4.85829960  3177 -6.283186   1697 -2.750716   7649   1.932303
## 644       0.52224371  3085 -2.895814   1844   8.662345   7103  -7.138188
## 645       1.69328459  3217  4.278768   1788 -3.036876   7615   7.208222
## 646      -3.74645222  3032 -5.750699   1739 -2.740492   8031   5.462902
## 647      -0.09828976  3095  2.077836   1669 -4.025302   7508  -6.512265
## 648       3.34513971  3266  5.525040   1635 -2.037148   7740   3.090037
## 649              NA    NA      NA      NA      NA      NA      NA
## 650              NA    NA      NA      NA      NA      NA      NA

```

```

# export data to csv
write.csv(FinalData, "Datasets/FinalData.csv", row.names = F)

```

Key Dates in Final Dataset:

- Claims data begins in 1971 and ends 2/2025
- Laus data begins in 1976 and ends 12/2024
- SP500 data begins in 1980 and ends 3/2025
- FRED data begins in 2000 and ends 12/2024
- SP500 Indices begins in 2009

Also, LFPR dates back to 1948, but the FRED datasets goes back to 2000. Therefore if it needs to be used then it can be loaded in separately.